

PROBLEM SOLVING AND TESTING USING JAVA

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Week1 questions :

Task 1. Access and print the element at a given index in an array.

Task 2. Search for a given element in a sorted array using Binary Search.

Task 3. Find the maximum element in an array of n integers.

Task 4. Given an array of integers and a positive integer K, write a program to find: The Kth smallest element.

Task 5. Print all possible pairs of elements from an array of size n.

Task 6: digitSum opt: sum of even or odd digits.

Task 7:Nth Fibonacci.

Task 8: Is Palindrome Number?

Task 1:

Program:

```
import java.util.stream.IntStream;

public class Task1 {

    public static void main(String[] args) {

        int[] arr = {10, 20, 30, 40, 50};

        int index = 2;

        IntStream.range(0, arr.length)

            .filter(i -> i == index)

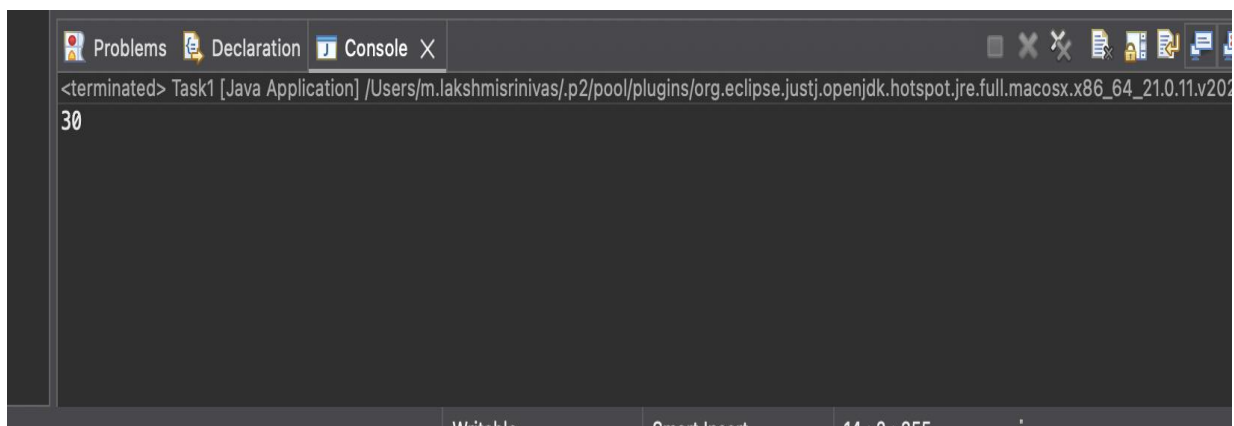
            .map(i -> arr[i])

            .forEach(System.out::println);

    }

}
```

Output:

A screenshot of the Eclipse IDE's console window. The window has tabs for 'Problems', 'Declaration', and 'Console'. The 'Console' tab is active, showing the output of the Java application. The output text is '<terminated> Task1 [Java Application] /Users/m.lakshmisrinivas/.p2/pool/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.x86_64_21.0.11.v2023-01-17-10.00/bin/java -Xmx1024m -Djava.awt.headless=true -Djava.class.path=.:./bin Task1 30'. The number '30' is printed on the first line of the console output, which corresponds to the element at index 2 of the array {10, 20, 30, 40, 50}. The status bar at the bottom of the IDE shows 'Writable', 'Smart Insert', and '14 x 2 = 255'.

Task 2:

Program:

```
interface BinarySearch {  
    int search(int[] arr, int low, int high, int key);  
}  
  
public class Task2 {  
    public static void main(String[] args) {  
        int[] arr = {10, 20, 30, 40, 50, 60, 70};  
        int key = 40;  
        BinarySearch bs = new BinarySearch() {  
            @Override  
            public int search(int[] arr, int low, int high, int key) {  
                BinarySearch lambda = (a, l, h, k) -> search(a, l, h,  
k);  
                if (low > high)  
                    return -1;  
                int mid = (low + high) / 2;  
                if (arr[mid] == key)  
                    return mid;  
                else if (key < arr[mid])  
                    return lambda.search(arr, low, mid - 1, key);
```

```
        else

            return lambda.search(arr, mid + 1, high, key);

    }

};

int index = bs.search(arr, 0, arr.length - 1, key);

if (index != -1)

    System.out.println("Element found at index: " +
index);

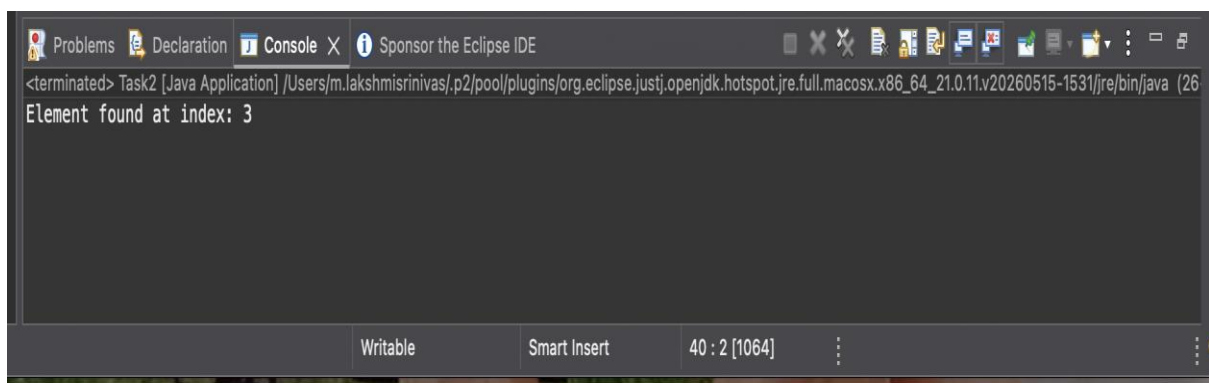
else

    System.out.println("Element not found");

}

}
```

Output:



The screenshot shows the Eclipse IDE interface. The top toolbar includes icons for Problems, Declaration, Console, and a link to 'Sponsor the Eclipse IDE'. The Console tab is active, displaying the output of a Java application. The output text is: <terminated> Task2 [Java Application] /Users/m.lakshmisrinivas/.p2/pool/plugins/org.eclipse.justi.openjdk.hotspot.jre.full.macosx.x86_64_21.0.11.v20260515-1531/jre/bin/java (26 Element found at index: 3. The bottom status bar shows 'Writable', 'Smart Insert', and '40 : 2 [1064]'.

```
<terminated> Task2 [Java Application] /Users/m.lakshmisrinivas/.p2/pool/plugins/org.eclipse.justi.openjdk.hotspot.jre.full.macosx.x86_64_21.0.11.v20260515-1531/jre/bin/java (26
Element found at index: 3
```

Task 3:

Program:

```
import java.util.Arrays;

public class Task3 {

    public static void main(String[] args) {

        int[] arr = {12, 45, 7, 89, 34, 56};

        int max = Arrays.stream(arr)

            .reduce((a, b) -> a > b ? a : b)

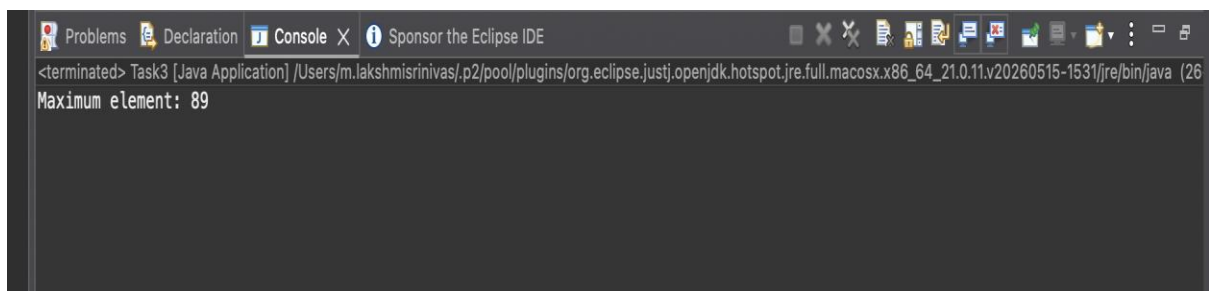
            .orElseThrow();

        System.out.println("Maximum element: " + max);

    }

}
```

Output:

A screenshot of the Eclipse IDE's console window. The window has a dark background and a light-colored title bar. The title bar contains several icons and text: 'Problems', 'Declaration', 'Console X', and 'Sponsor the Eclipse IDE'. Below the title bar, the console output is displayed in a monospaced font. The first line of output is '<terminated> Task3 [Java Application] /Users/m.lakshmisrinivas/.p2/pool/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.x86_64_21.0.11.v20260515-1531/jre/bin/java (26'. The second line of output is 'Maximum element: 89'. The console window is currently empty except for these two lines of text.

```
<terminated> Task3 [Java Application] /Users/m.lakshmisrinivas/.p2/pool/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.x86_64_21.0.11.v20260515-1531/jre/bin/java (26
Maximum element: 89
```

Task 4:

Program:

```
import java.util.Arrays;

public class Task4{

    public static void main(String[] args) {

        int[] arr = {7, 10, 4, 3, 20, 15};

        int k = 3;

        int kthSmallest = Arrays.stream(arr)

            .sorted()

            .skip(k - 1)

            .findFirst()

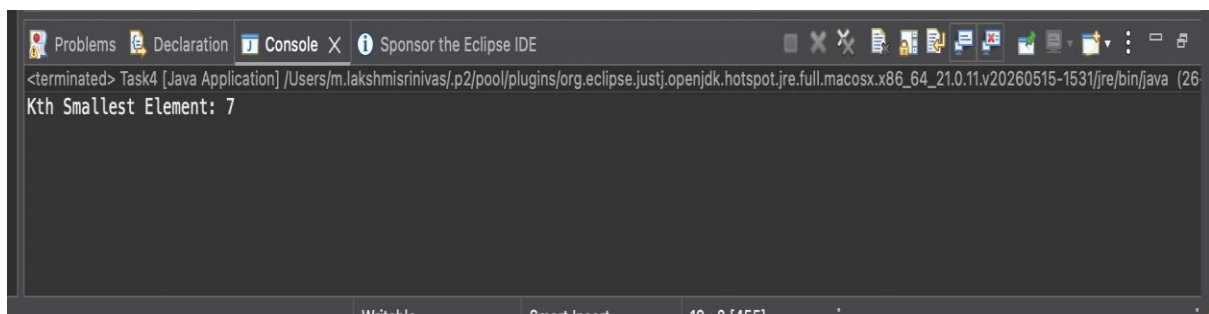
            .orElseThrow();

        System.out.println("Kth Smallest Element: " +
kthSmallest);

    }

}
```

Output:

A screenshot of the Eclipse IDE's console window. The console shows the output of the Java application: "Kth Smallest Element: 7". The console title bar indicates the application is "Task4 [Java Application]" and the path is "/Users/m.lakshminivas/.p2/pool/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.x86_64_21.0.11.v20260515-1531/jre/bin/java (26". The console is currently empty except for the output line.

```
<terminated> Task4 [Java Application] /Users/m.lakshminivas/.p2/pool/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.x86_64_21.0.11.v20260515-1531/jre/bin/java (26
Kth Smallest Element: 7
```

Task 5:

Program:

```
import java.util.stream.IntStream;

public class Task5 {

    public static void main(String[] args) {

        int[] arr = {1, 2, 3, 4};

        IntStream.range(0, arr.length)

            .forEach(i ->

                IntStream.range(i + 1, arr.length)

                    .forEach(j ->

                        System.out.println("(" + arr[i] + ", " + arr[j] +

                            ")")

                    )

            );

    }

}
```

Output :



```
<terminated> Task5 [Java Application] /Users/m.lakshminivas/p2/pool/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.x86_64_21.0.11.v20260515-1531/jre/
(1, 2)
(1, 3)
(1, 4)
(2, 3)
(2, 4)
(3, 4)
```

Task 6:

Program:

```
import java.util.stream.IntStream;

public class Task6 {

    public static void main(String[] args) {

        int number = 123456;

        String option = "even"; // Change to "odd" for sum of
odd digits

        int sum = String.valueOf(number)

            .chars()

            .map(c -> c - '0')

            .filter(d -> option.equalsIgnoreCase("even") ? d

% 2 == 0 : d % 2 != 0)

            .sum();

        System.out.println("Sum of " + option + " digits = " +
sum);

    }

}
```

Output:



```
<terminated> Task6 [Java Application] /Users/m.lakshminivas/p2/pool/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.x86_64_21.0.11.v20260515-1531/jre/bin/java (26
Sum of even digits = 12
```


Task 7:

Program:

```
import java.util.function.IntUnaryOperator;

public class Task7{

    public static void main(String[] args) {

        int n = 10; // Find the 10th Fibonacci number

        IntUnaryOperator fib = new IntUnaryOperator() {

            @Override

            public int applyAsInt(int num) {

                if (num <= 1)

                    return num;

                return this.applyAsInt(num - 1) +

this.applyAsInt(num - 2);

            }

        };

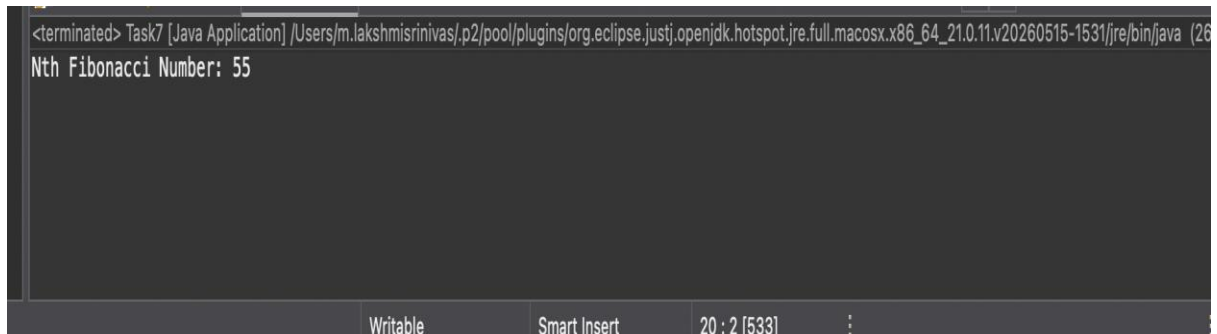
        System.out.println("Nth Fibonacci Number: " +

fib.applyAsInt(n));

    }

}
```

Output:

A screenshot of a Java application window titled '<terminated> Task7 [Java Application]'. The window shows the output 'Nth Fibonacci Number: 55'. The status bar at the bottom indicates 'Writable', 'Smart Insert', and '20 : 2 [533]'.

```
<terminated> Task7 [Java Application] /Users/m.lakshmisrinivas/.p2/pool/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.x86_64_21.0.11.v20260515-1531/jre/bin/java (26
Nth Fibonacci Number: 55
Writable Smart Insert 20 : 2 [533]
```

Task 8:

Program:

```
import java.util.function.Predicate;

public class Task8{

    public static void main(String[] args) {

        int number = 121;

        Predicate<Integer> isPalindrome = n -> {

            int original = n;

            int reverse = 0;

            while (n > 0) {

                reverse = reverse * 10 + (n % 10);

                n /= 10;

            }

            return original == reverse;

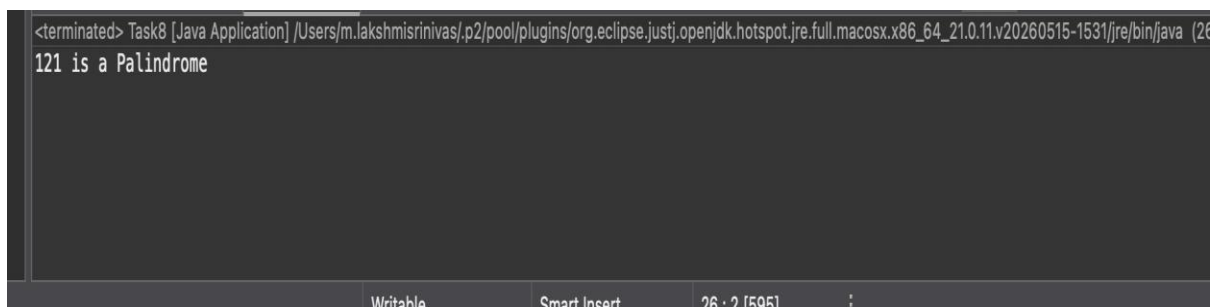
        }

    }

}
```

```
};  
if (isPalindrome.test(number))  
    System.out.println(number + " is a Palindrome");  
else  
    System.out.println(number + " is not a Palindrome");  
}  
}
```

Output:



The screenshot shows a Java application window titled "<terminated> Task8 [Java Application]". The window contains a text area with the output "121 is a Palindrome". The status bar at the bottom indicates "Writable", "Smart Insert", and "26 : 2 (595)".

Task 9:

Program:

```
import java.util.function.BiFunction;

public class Task9{

    public static void main(String[] args) {

        int num1 = 1234;

        int num2 = 5678;

        BiFunction<Integer, Integer, Integer> sumLastDigits =

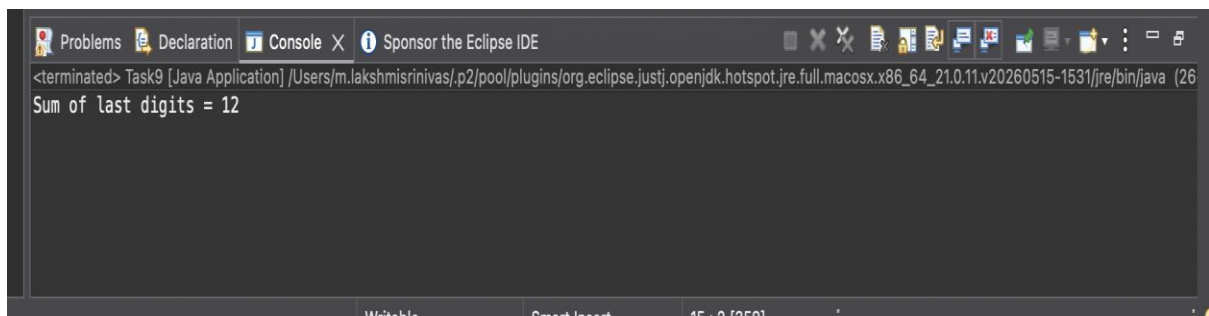
            (a, b) -> (a % 10) + (b % 10);

        System.out.println("Sum of last digits = " +
sumLastDigits.apply(num1, num2));

    }

}
```

Output:

A screenshot of the Eclipse IDE's console window. The console shows the output of the Java application: "Sum of last digits = 12". The window title is "Task9 [Java Application]". The console also displays the path to the Java runtime: "/Users/m.lakshmisrinivas/.p2/pool/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.x86_64_21.0.11.v20260515-1531/jre/bin/java (26". The status bar at the bottom indicates "Writable", "Smart Insert", and "15 · 2 [359]".

```
<terminated> Task9 [Java Application] /Users/m.lakshmisrinivas/.p2/pool/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.x86_64_21.0.11.v20260515-1531/jre/bin/java (26
Sum of last digits = 12
```